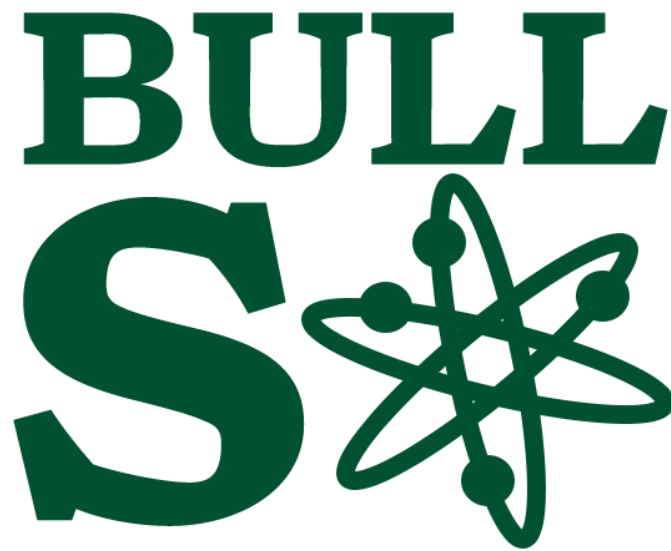


Remote Sensing B/C



Bull Science Olympiad at the University of South Florida

Saturday, January 24th, 2026

Team Number: _____

Team Name: _____

Participant Names: _____

Total Score: _____/120

Rank: _____

Section 1: Fundamentals of Remote Sensing (Each MCQ is 1 pt)

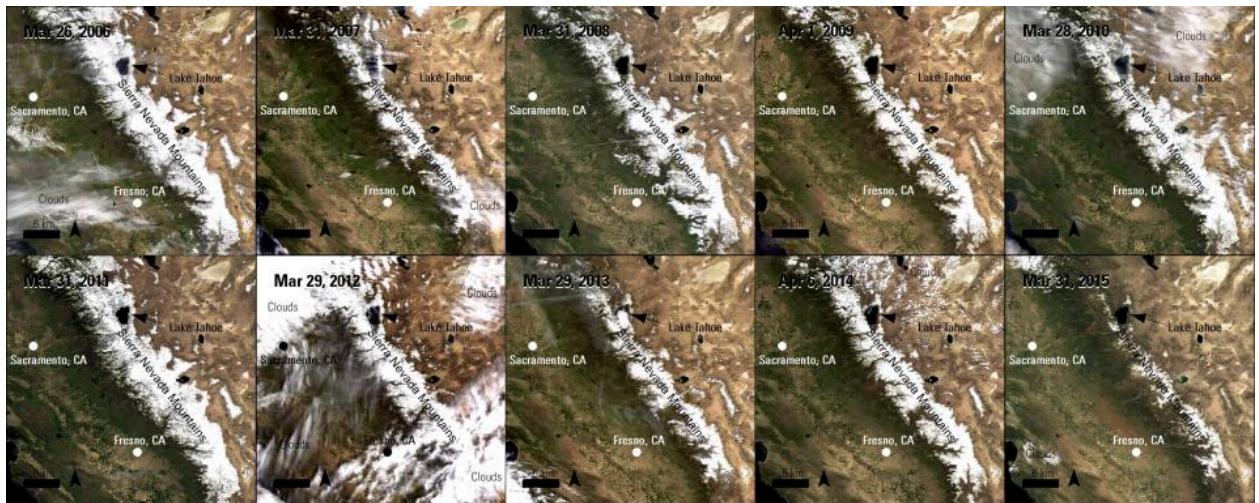
1. Remote sensing is best defined as the acquisition of information about an object or area by:
 - A. Direct physical contact
 - B. Ground-based surveys only
 - C. Measuring reflected or emitted energy from a distance
 - D. Laboratory experimentation
2. Which portion of the electromagnetic spectrum is most commonly used for Earth observation?
 - A. Gamma rays
 - B. Visible and infrared wavelengths
 - C. X-rays
 - D. Radio waves only
3. Passive remote sensing systems rely on:
 - A. Energy emitted from the sensor
 - B. Reflected or emitted natural energy
 - C. Mechanical waves
 - D. Magnetic fields
4. Active remote sensing differs from passive sensing because it:
 - A. Cannot operate at night
 - B. Uses naturally occurring radiation
 - C. Emits its own energy toward a target
 - D. Measures temperature only
5. Which quantity describes the amount of energy arriving at a surface per unit area?
 - A. Radiance
 - B. Irradiance
 - C. Reflectance
 - D. Albedo
6. Reflectance is best described as the ratio of:
 - A. Emitted to absorbed energy
 - B. Reflected to incident energy
 - C. Transmitted to emitted energy
 - D. Absorbed to transmitted energy
7. Which surface would generally have the highest albedo?
 - A. Dense forest
 - B. Open ocean
 - C. Fresh snow
 - D. Dark asphalt
8. The electromagnetic spectrum is organized primarily by differences in:
 - A. Energy source

- B. Frequency and wavelength
 - C. Direction of travel
 - D. Polarization only
9. Which interaction occurs when radiation passes through a material without significant energy loss?
- A. Absorption
 - B. Reflection
 - C. Transmission
 - D. Emission
10. Atmospheric scattering most strongly affects which wavelengths?
- A. Longwave infrared
 - B. Microwave
 - C. Shorter visible wavelengths
 - D. Thermal infrared
-

Section 2: Sensors & Platforms

11. A satellite is considered sun-synchronous if it:
- A. Remains stationary over one point on Earth
 - B. Passes over locations at the same local solar time
 - C. Operates only during daylight
 - D. Has a circular equatorial orbit
12. Which platform provides the highest spatial resolution?
- A. Geostationary satellite
 - B. Polar-orbiting satellite
 - C. Aircraft-based sensor
 - D. Weather balloon
13. Spatial resolution refers to:
- A. Number of spectral bands
 - B. Time between observations
 - C. Smallest object detectable by a sensor
 - D. Accuracy of radiometric measurements
14. Temporal resolution describes a sensor's ability to:
- A. Detect small features
 - B. Measure wavelength ranges
 - C. Revisit the same location over time
 - D. Penetrate cloud cover
15. Which sensor characteristic relates to the number of wavelength bands measured?
- A. Spatial resolution

- B. Temporal resolution
 - C. Spectral resolution
 - D. Radiometric resolution
16. Radiometric resolution is determined by the sensor's ability to:
- A. Detect fine spatial details
 - B. Distinguish small differences in energy
 - C. Capture multiple wavelengths
 - D. Revisit targets frequently
17. Observe this image and answer the following questions:



This image was taken in the Sierra Nevada Mountains between 2006 and 2015. What satellite took this image?

- A. MODIS Aqua
 - B. MODIS Terra
 - C. GOES
 - D. Suomi NPP
18. Which type of sensor is most useful for imaging Earth's surface through clouds?
- A. Optical
 - B. Thermal infrared
 - C. Microwave radar
 - D. Ultraviolet
19. Pushbroom scanners differ from whiskbroom scanners in that pushbroom scanners:
- A. Use moving mirrors
 - B. Measure one pixel at a time
 - C. Use linear detector arrays
 - D. Cannot be satellite-mounted
20. Which platform is best suited for continuous weather monitoring?
- A. Aircraft
 - B. Polar-orbiting satellite

- C. Geostationary satellite
 - D. Ground station
-

Section 3: Spectral Signatures & Surface Features

- 21. A spectral signature describes how a material:
 - A. Emits heat only
 - B. Reflects and absorbs energy across wavelengths
 - C. Changes shape with temperature
 - D. Moves relative to the sensor
- 22. Healthy vegetation strongly reflects which wavelength region?
 - A. Blue light
 - B. Red light
 - C. Near-infrared
 - D. Thermal infrared
- 23. Water typically appears dark in near-infrared imagery because it:
 - A. Reflects most incoming energy
 - B. Strongly absorbs NIR radiation
 - C. Emits visible light
 - D. Scatters microwaves
- 24. Bare soil reflectance generally increases as wavelength:
 - A. Decreases
 - B. Remains constant
 - C. Increases
 - D. Reaches ultraviolet
- 25. A false-color composite that assigns near-infrared to red is useful for identifying:
 - A. Urban heat islands
 - B. Vegetation health
 - C. Cloud height
 - D. Ocean salinity
- 26. Which land cover type typically shows high reflectance in visible wavelengths but low reflectance in NIR?
 - A. Healthy crops
 - B. Forest canopy
 - C. Snow-covered ground
 - D. Open water
- 27. Spectral confusion occurs when:
 - A. Sensors fail completely
 - B. Two materials have similar reflectance patterns

- C. Images are taken at night
 - D. Data are overcorrected
28. Which factor most directly affects a material's spectral signature?
- A. Sensor altitude
 - B. Surface composition
 - C. Orbital inclination
 - D. Pixel size
29. Which of the following satellites most likely took this image?



- A. AURA
 - B. TOPEX-Poseidon
 - C. Suomi NPP
 - D. GOES-16
30. Vegetation stress is most effectively detected using changes in:
- A. Blue reflectance
 - B. Near-infrared reflectance
 - C. Microwave backscatter
 - D. Ultraviolet emission

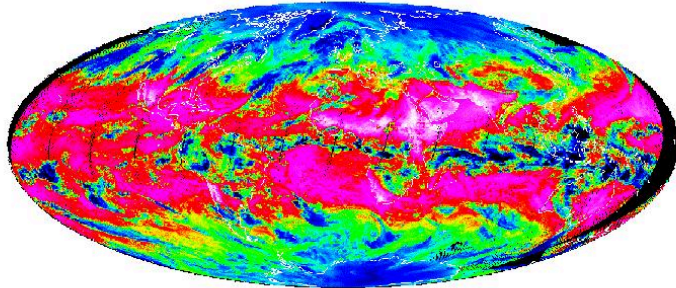
Section 4: Energy Balance & Climate Applications

31. Earth's energy balance is primarily between incoming solar radiation and:
- A. Geothermal energy

- B. Reflected shortwave radiation
 - C. Outgoing longwave radiation
 - D. Atmospheric pressure
32. Thermal infrared sensors measure energy that is primarily:
- A. Reflected sunlight
 - B. Emitted by Earth's surface
 - C. Scattered by clouds
 - D. Absorbed by vegetation
33. Surface temperature retrieval relies on measurements in which region?
- A. Visible
 - B. Near-infrared
 - C. Thermal infrared
 - D. Microwave only
34. An increase in surface albedo would most likely result in:
- A. Higher surface temperatures
 - B. Lower reflected energy
 - C. Reduced absorption of solar energy
 - D. Increased greenhouse effect
35. Which remote sensing product is most useful for monitoring drought?
- A. Sea surface height
 - B. Vegetation indices
 - C. Magnetic field strength
 - D. Topographic maps
36. The Normalized Difference Vegetation Index (NDVI) compares reflectance in the:
- A. Blue and green bands
 - B. Red and near-infrared bands
 - C. Thermal and microwave bands
 - D. Ultraviolet and visible bands
37. Clouds generally appear bright in visible imagery because they:
- A. Emit thermal radiation
 - B. Absorb infrared radiation
 - C. Strongly reflect sunlight
 - D. Transmit most energy
38. Long-term satellite observations are especially useful for studying:
- A. Daily weather events only
 - B. Short-term storms
 - C. Climate trends over time
 - D. Local soil chemistry
39. Which factor most influences the amount of solar energy reaching Earth's surface?
- A. Orbital velocity

- B. Atmospheric composition
 - C. Sensor resolution
 - D. Pixel size
40. The Red/Yellow areas within this image indicate what type of flux:

Emitted Thermal Flux
Measured By CERES on Terra
Preliminary Data: February 26, 2000



NASA Langley Research Center / Atmospheric Sciences



- A. Net incoming flux (indicating heating)
- B. Net incoming flux (indicating cooling)
- C. Net outgoing flux (indicating heating)
- D. Net outgoing flux (indicating cooling)

Section 5: Data Processing & Applications

41. Radiometric correction is performed to:
- A. Improve spatial resolution
 - B. Remove sensor and atmospheric effects
 - C. Increase revisit frequency
 - D. Change spectral bands
42. Geometric correction ensures that:
- A. Pixel values are normalized
 - B. Images align with real-world coordinates
 - C. Noise is eliminated
 - D. Resolution is enhanced
43. Classification of remote sensing data involves:
- A. Sorting pixels into categories
 - B. Removing atmospheric interference

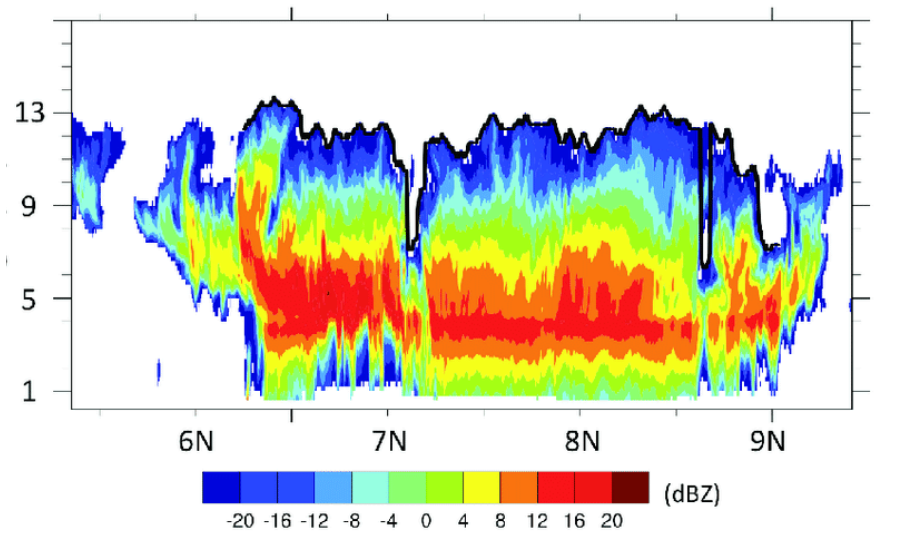
- C. Increasing spectral resolution
 - D. Measuring orbital height
44. Supervised classification requires:
- A. No prior knowledge of the area
 - B. Training data provided by the analyst
 - C. Automatic clustering only
 - D. Thermal imagery exclusively
45. Unsupervised classification differs because it:
- A. Uses labeled training samples
 - B. Requires ground truth data
 - C. Groups pixels based on spectral similarity
 - D. Produces no output map
46. Ground truth data are collected to:
- A. Replace satellite imagery
 - B. Validate and interpret remote sensing results
 - C. Improve sensor calibration only
 - D. Increase image resolution
47. Change detection analysis compares images from:
- A. Different sensors only
 - B. Different times
 - C. Different wavelengths
 - D. Identical conditions
48. Remote sensing is especially useful in disaster response because it:
- A. Requires no calibration
 - B. Provides rapid, large-area coverage
 - C. Eliminates field surveys
 - D. Operates only in clear weather
49. Which application most directly uses microwave remote sensing?
- A. Vegetation health mapping
 - B. Soil moisture estimation
 - C. Cloud color analysis
 - D. Photosynthetic rate calculation
50. One major advantage of satellite remote sensing is its ability to:
- A. Measure properties at a single point only
 - B. Replace all in situ measurements
 - C. Observe large areas consistently over time
 - D. Operate without data processing

Section 6: Radiative Forcing and Spectroscopy (18 Points)

1. (3 pts) Define **Radiative Forcing**. Is a positive RF associated with warming or cooling?
2. (4 pts) Contrast **Mie Scattering** and **Rayleigh Scattering** in terms of particle size relative to wavelength ($x = 2\pi r/\lambda$). Provide one real-world example of an atmospheric constituent that causes each.
3. (3 pts) How does the presence of **Sulfate Aerosols** in the stratosphere after a major volcanic eruption affect the Earth's energy balance? Mention the specific effect on albedo.
4. (4 pts) Calculate the Blackbody Emission (E) for a surface at 300 K using the Stefan-Boltzmann Law ($\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$). If the emissivity (ϵ) is 0.9, what is the actual flux?
5. (4 pts) Explain the **GWP** (Global Warming Potential) of Methane relative to CO₂ over a 20-year vs. a 100-year timeframe. Why does this ratio change?

Section 7: Clouds and Precipitation (15 Points)

1. (3 pts) Compare **Cirrus** and **Stratus** clouds in terms of their net effect on the Earth's radiation budget (consider both albedo and the greenhouse effect).



2. (4 pts) This is a CloudSat profile. In general, what does the "Reflectivity (dBZ)" scale tell us about the size and concentration of particles in a cloud? What is the y axis label?
3. (4 pts) Explain how the **GPM (Global Precipitation Measurement)** mission uses a "Dual-frequency Precipitation Radar" (DPR) to distinguish between liquid rain and frozen snow.
4. (4 pts) Define **Aerosol Optical Depth (AOD)**. How does a high AOD affect the amount of solar radiation reaching the surface?

Section 8: Orbital Mechanics and Resolution (12 Points)

1. (4 pts) A satellite is in a circular orbit at an altitude of 700 km. Calculate its **orbital period** in minutes. (Earth Radius = 6,371 km; $G = 6.67 \times 10^{-11}$; $M_E = 5.97 \times 10^{24} \text{ kg}$).
2. (3 pts) Why are **Sun-synchronous orbits** preferred for "Change Detection" studies (e.g., monitoring deforestation over several years)?

3. (3 pts) If a sensor has a **revisit period** of 16 days but a swath width of 185 km, how would decreasing the altitude affect its revisit period?
 4. (2 pts) Distinguish between **Spatial Resolution** and **Temporal Resolution**.
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Section 9: Greenhouse Gas Sinks and Cycles (19 Points)

1. (4 pts) Describe the **Biological Pump** in the ocean. How does it act as a carbon sink?
 2. (3 pts) What is the role of the **Brewer-Dobson Circulation** in distributing trace gases like Ozone and Nitrous Oxide?
 3. (4 pts) Discuss the impact of **Ocean Acidification** (caused by CO₂ absorption) on marine calcifiers. How might this be observed indirectly via remote sensing?
 4. (4 pts) Calculate the expected temperature increase (ΔT) if the CO₂ concentration doubles, given a climate sensitivity λ of 0.8 K / (W/m²) and the formula for forcing: $\Delta F = 5.35 \ln(C/C_0)$.
 5. (4 pts) Identify two human activities that act as major sources of **Methane** (CH₄).
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Section 10 Energy Balance Modeling (6 Points)

1. (6 pts) Solve for the **Equilibrium Temperature** of the Earth's surface if the Albedo (α) is 0.3 and the Solar Constant (S_0) is 1361 W/m².